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Comparison of a novel patented method of measuring Blood pressure (Plethysmometry) with Sphygmomanometry

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Abstract

Background: We have developed a patented method that acquires photoplethysmograms (PPG) at multiple cuff-pressure levels to estimate Blood pressure (BP). The method is referred to as plethysmometry. Validation of new methods for measuring BP involves comparison with Sphygmomanometry as standard.

Objective: To compare the new method of BP measurement (plethysmometry) with Sphygmomanometry guided by international protocols for validation and to evaluate diagnostic agreement for hypertension (HT).

Methods: Ninety-two adults underwent four video/audio-documented sphygmomanometric readings (two before and two after plethysmometry). The lowest and highest of the four estimates were taken as lower and upper limits of systolic and diastolic BP reported by sphygmomanometry. The values were averaged to get average systolic and diastolic pressures for each individual.

Plethysmometry involves recording of 3 signals; cuff-pressure, and PPG from fingers of both cuffed and uncuffed arms. A brief ramp-inflation of cuff-pressure to the point at which the cuffed-arm pulse disappears provides a preliminary systolic estimate. 16 cuff-pressure steps (ranging from above the preliminary systolic estimate to 30 mmHg) were then applied, holding cuff-pressure at each level for 10 seconds. From the relationship of pulse amplitude to cuff-pressure, lower and upper limits of systolic and diastolic pressures were derived. Average systolic and diastolic pressures and average pulse pressure were calculated. Mean arterial pressure (MAP) was obtained from the relationship between cuff-pressure and cuff-pressure oscillation amplitude. The average pressures from both methods were compared as per AAMI criterion-1 and BHS grading scheme.

A new scheme of BP classification with Plethysmometry was designed. Normative data was obtained from a subset of 77 individuals who did not have previous history of HT and whose sphygmomanometric pressures were below 140/90 mmHg. Based on the relationship of the systolic, diastolic, mean arterial and pulse pressures of the subject to the normative pressures, BP was classified as HT or normotension (NT).

Results: Plethysmometry gives two sets of arterial pressures, one proximal to the arm-cuff (closer to central pressures) and the other, in the arterial segment below or distal to the cuff (peripheral). Comparing average systolic and diastolic pressures between plethysmometry (central) and sphygmomanometry, bias(SD) was 1.18(6.66) mmHg(systolic) and -4.17(9.07) mmHg (diastolic); AAMI criterion-1 was met for central systolic but not for diastolic; criterion-2 was not applicable as there was only a single plethysmometric measurement per subject. BHS grading criteria also were met for central systolic but not for diastolic pressure. When HT diagnosis by plethysmometry with the new scheme (incorporating systolic, diastolic, mean arterial and pulse pressure) was compared with sphygmomanometry, sensitivity and specificity were more than 0.9.

Conclusions: Plethysmometry provides objective ranges of systolic and diastolic pressures and MAP from a single recording, verifiable by traces of recorded parameters. It is equivalent to multiple sphygmomanometric measurements done under ideal conditions in terms of high diagnostic accuracy for HT as well as agreement between systolic pressures.

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The consistent negative diastolic bias in plethysmometry may be due to the well-documented over-estimation of diastolic pressure by sphygmomanometry. Plethysmometry is a good tool for clinic BP assessment and can detect masked hypertension.

Keywords: Plethysmometry, Sphygmomanometry, Hypertension, BP variability

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Introduction: Currently, two cuff-based methods are used clinically for blood pressure (BP) measurement—Sphygmomanometry and Oscillometry. Both employ an inflatable cuff encircling the arm, which can be pressurized, transmitting the pressure to the underlying artery.

In sphygmomanometry, when the cuff pressure exceeds the true systolic pressure in the underlying artery, arterial flow ceases and no sounds are heard through a stethoscope placed over the cubital fossa. When cuff pressure falls below the true diastolic pressure, laminar flow resumes and the sounds again disappear.

When cuff pressure lies between systolic and diastolic levels, intermittent flow produces vibrations in the arterial wall, generating the Korotkoff sounds. As the cuff is deflated from a high pressure, the cuff-pressure at the first appearance of these sounds indicates systolic pressure, and the cuff-pressure at their disappearance indicates diastolic pressure. The sounds arise from complex interactions between intermittent flow and arterial wall vibrations (1).

In Oscillometry, cuff pressure oscillates with each pulse. The oscillation amplitude is maximal when cuff pressure approximates the mean arterial pressure (MAP). Systolic and diastolic pressures are then estimated from empirical ratios of oscillation amplitude relative to this maximum, or from the shape of the oscillation envelope. The exact algorithms are proprietary to manufacturers.

Sphygmomanometry remains the reference standard against which new devices are validated, as recommended by the collaboration statement of Association for the Advancement of Medical Instrumentation (AAMI), European Hypertension Society (EHS) and International Organization for standardization (2).

However, recent systematic reviews have highlighted inaccuracies in both systolic and diastolic readings obtained by sphygmomanometry (3). Because Oscillometric devices are calibrated to sphygmomanometric readings, such errors inevitably propagate into Oscillometric measurements.

Several investigators have compared intra-arterial radial or aortic pressures with cuff-based estimates. Bui *et al.* and Picone *et al.* have emphasized that cuff BP was originally intended to approximate aortic (central) pressure, since the target organs affected by hypertension experience aortic rather than peripheral pressures (3, 4).

Bui *et al.* (4) conducted a meta-analysis of 2013 participants from 31 studies and found that Oscillometric devices overestimated systolic BP in children but underestimated it in older adults, while diastolic BP was overestimated, with the magnitude of error increasing with age. Such systematic age-related bias has major implications for diagnostic thresholds and treatment decisions.

Picone et al. (5) analysed five studies (n = 262) comparing Oscillometric and intra-aortic pressures during cardiac catheterization and found significant differences in mean arterial pressure (MAP), indicating that Oscillometric MAP cannot be assumed equivalent to invasive MAP. They concluded that discrepancies in systolic and diastolic pressures measured by Oscillometric methods arise largely from MAP estimation errors.

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The meta-analysis (3) comparing cuff-based and invasive measurements confirmed that invasive brachial systolic pressure exceeds invasive aortic pressure, while diastolic pressures differ little between central and peripheral sites. However, individual variability was high. Across studies, cuff methods significantly underestimated intra-arterial brachial systolic BP and overestimated diastolic BP ($p < 0.0001$). When compared with intra-aortic pressure, cuff systolic BP showed mixed over- and under-estimation, averaging an absolute difference of 8 mmHg, while diastolic BP was consistently over-estimated.

Clinically, what matters is not exact numeric agreement but whether BP classification by cuff methods aligns with invasive reference categories. The meta-analysis (3) concluded that cuff readings below 120/80 mmHg or above 160/100 mmHg provide reasonable confidence, but classification accuracy is poor in the intermediate “pre-hypertensive to stage I” range (140/90–160/100 mmHg).

A further under-recognized feature of BP is its short-term variability over minutes, driven by Traube (respiratory) and Mayer (low-frequency) waves. From intra-arterial recordings in 51 hemodynamically stable ICU patients, we previously reported variability of 21 ± 9 mmHg for systolic and 14 ± 5 mmHg for diastolic pressure (6). No cost-effective clinical device currently quantifies this short-term BP variability (BPV).

In this context, we have developed a novel cuff-based BP device (patents 7–9) that provides not just single-point values but ranges of systolic and diastolic pressure over a few minutes. This variability may be referred to as ‘Ultra short-term BPV. As the objective of this paper is to compare the new method with a reference, the variability component is not discussed further.

The device measures pulse amplitudes at varying cuff pressures using a plethysmograph and estimates systolic and diastolic pressure limits from the relationship of pulse amplitude to cuff-pressure. The new patented method is referred to as 'Plethysmometry' hereafter. The method reports Mean Arterial Pressure (MAP) too from considerations of ‘cuff-pressure oscillation amplitudes’.

In this study, we compare systolic and diastolic ranges obtained by a single plethysmometric recording in 92 individuals with ranges derived from four video- and audio-documented sphygmomanometric measurements reported by 2 independent observers. Hypertension classification from the plethysmometric method using a scoring system is evaluated against the sphygmomanometric reference.

Methods: The study was approved by the institutional review board. 92 subjects were enrolled in the study after obtaining informed consent. Their blood pressure was assessed with both sphygmomanometry and plethysmometry.

Sphygmomanometry: Each participant underwent four mercury sphygmomanometric measurements in a sound-proof room—two before and two after the plethysmometric recording. The mercury column was video-recorded and Korotkoff sounds were captured simultaneously. Two observers independently reviewed the synchronized video–audio recordings. If their systolic or diastolic readings differed by > 2 mmHg, a third observer served as arbiter. Values from the two observers were averaged for each reading. The

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lowest and highest of the four averaged readings were designated as the minimum and maximum systolic (or diastolic) pressures, respectively.

Plethysmometry: The patented plethysmometric system consists of an inflatable arm cuff and two photoplethysmographic (PPG) sensors—one on a finger of the cuffed arm and a reference sensor on the corresponding finger of the contralateral arm. The cuff is inflated and deflated by a digitally controlled pump while cuff pressure is monitored by an integrated pressure transducer. Cuff pressure and the two PPG waveforms are recorded simultaneously through three channels at a sampling frequency of 500 samples per second using the device's built-in data recorder. The recorded data are subsequently analysed to derive the upper and lower limits of systolic and diastolic BP over the recording period.

Cuff Pressure Protocols

Ramp Protocol: Initially, cuff pressure is increased in a continuous ramp, similar to conventional methods, to identify the pressure at which the pulse in the cuffed arm disappears. This provides a preliminary or approximate estimate of the systolic pressure.

Step Protocol: The pressure range between 30 mmHg and (systolic estimate from ramp + 30 mmHg) is divided into 16 equal steps. The pump is programmed to first raise the cuff pressure to the maximum step, hold it for approximately 10 seconds, and then release pressure completely to allow full restoration of circulation. After an interval of 10 seconds, the next lower pressure level is applied, and the process is repeated for all 16 steps. Throughout this sequence, cuff pressure and the photoplethysmogram (PPG) signals from both the cuffed and uncuffed arms are digitized and recorded.

The 10-second duration for each pressure plateau allows acquisition of pulses spanning at least one complete respiratory cycle. When cuff pressure exceeds systolic pressure, the PPG signal in the cuffed arm is absent. When it falls below diastolic pressure, the PPG amplitude is fully restored. Between these limits, the PPG signal is present but of reduced amplitude. Under normal conditions, PPG amplitude is proportional to pulse pressure (systolic − diastolic). At cuff pressures higher than diastolic, the PPG amplitude reflects the difference between systolic and cuff pressure. PPG amplitude therefore varies within a pressure step because of: (a) fluctuations in systolic and diastolic pressures when cuff pressure is below diastolic, and (b) systolic variations alone when cuff pressure is above diastolic (assuming vascular resistance remains constant).

All recordings were analysed using custom MATLAB algorithms developed in our laboratory.

Results: The study subjects included 40 women and 52 men. Tables 1a and 1b provide details of the sample and anthropometric data.

Table 1a: *Subject profile in terms of hypertension:*

1	Total subjects recruited	92
2	Known hypertensives in the sample of 92	8
3	Subjects after exclusion of known hypertensives (92 - 8)	84
4	Subjects with hypertension from sphygmomanometry in the sample of 84	7
5	Subjects from whom normative mean taken for plethysmometry (84 - 7)	77

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Of the 8 subjects in the sample with a previous history of hypertension, 4 were on treatment.

Table 1b: *Anthropometric data of study subjects (n=92)*

n = 92	Age in years	Height in cm	Weight in kg	BMI	Mid arm circumference in cm
Mean ± SD	32 ± 11	166 ± 10	68 ± 14	24 ± 4	30 ± 3
median	29	168	67	24	30
Range	18 - 61	147 - 189	36 - 114	16 – 35	22 - 37

Sample Raw Data: Figure 1 shows representative raw data from one experiment. In the first four pressure plateaus (in the top panel), no pulse is detectable in the cuffed arm (middle panel). The cuff pressure of approximately 120 mmHg (arrow A, top panel) is the lowest pressure at which all pulses are absent (A in middle panel). Pulses first reappear at a cuff pressure of 111 mmHg (arrow B in top panel and B in middle panel). At 68 mmHg (arrow C in top panel), the PPG amplitude is fully restored (C in middle panel), indicating unobstructed flow in the cuffed arm.

The amplitude of the photoplethysmographic (PPG) signal from both arms was measured for every pulse, and the ratio of cuff PPG amplitude to reference PPG amplitude (PPG ratio) was computed. For each pressure plateau, the average cuff pressure was calculated and taken as the representative cuff pressure for that plateau, since cuff pressure exhibits small oscillations within each level. A plot of PPG ratio versus cuff pressure is shown in Figure 2.

Figure 1: *Sample raw data demonstrating simultaneous recording of Cuff-pressure (top panel), Pulse from cuffed arm (middle panel) and pulse from uncuffed arm (bottom panel).*

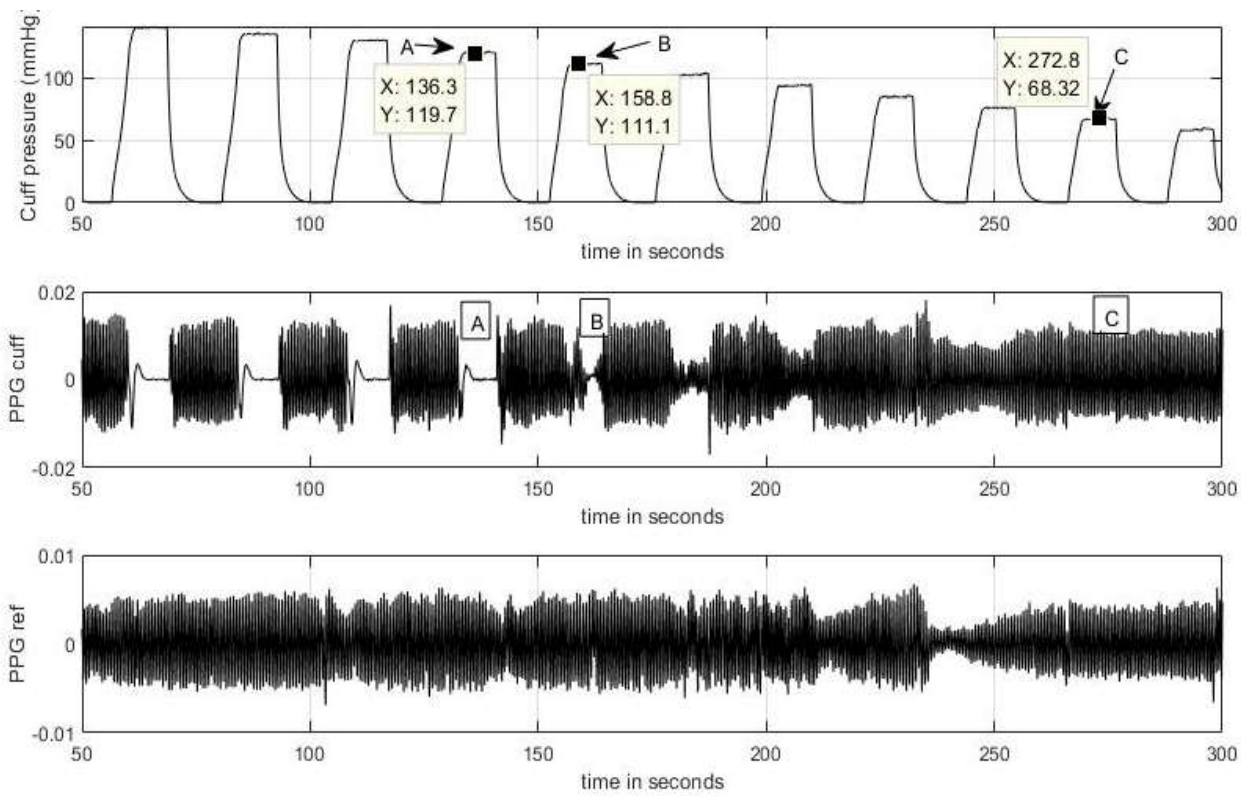

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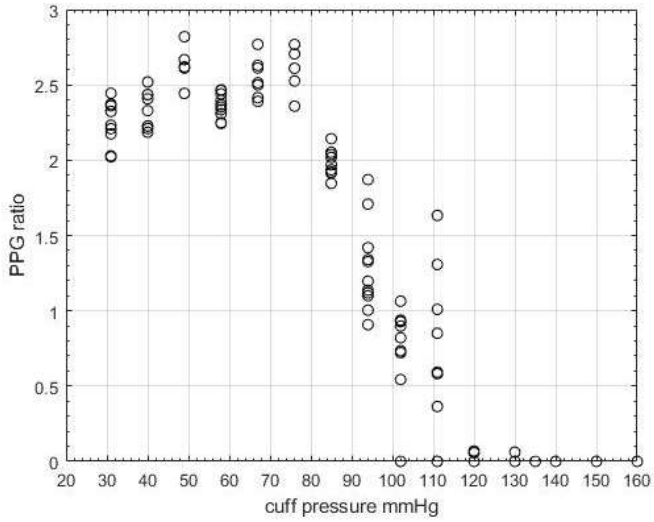
***Figure 2**: Sigmoid relationship between cuff-pressure and PPG ratio:*

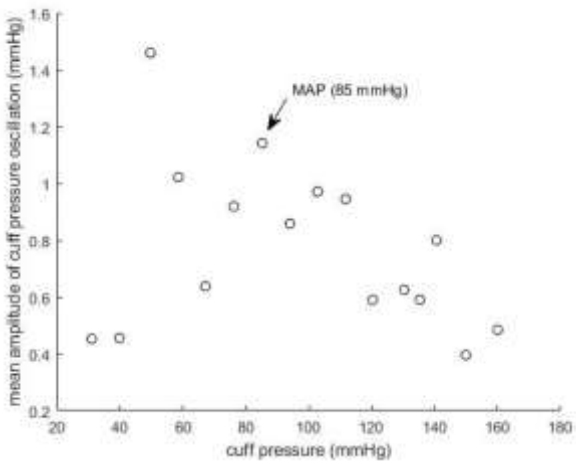


From the relationship shown in Figure 2, the lower and upper limits of systolic and diastolic pressures were derived algorithmically using curve-fitting procedures. The lower systolic pressure corresponds to the minimum cuff pressure at which at least one pulse is absent, indicating that the systolic pressure for that pulse is below the cuff pressure. The upper systolic pressure is defined as the minimum cuff pressure at which all pulses are absent. Similarly, the upper diastolic pressure is the maximum cuff pressure at which at least one pulse amplitude is fully restored—comparable to the unpressurized state—whereas the lower diastolic pressure is the cuff pressure at which all pulse amplitudes resemble those recorded without cuff inflation.

These pressures are referred to as central pressures, because the presence of a pulse in the cuffed arm during a pressure plateau implies that the pressure proximal to the cuff—say, the subclavian pressure, which is closer to aortic pressure—is higher than the cuff pressure.

The amplitude of cuff-pressure oscillations at each plateau was also quantified. Figure 3 illustrates the plot of mean oscillation amplitude versus cuff pressure. From this relationship, the mean arterial pressure (MAP) and the peripheral systolic and diastolic pressures (either within or distal to the occluded artery) were estimated using custom MATLAB algorithms.

***Figure 3:** Mean oscillation amplitude versus cuff pressure during a plateau*



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For representative recording from one subject shown in Figures 1–3, the corresponding estimates of central and peripheral pressures are given in Table 2.

Table 2. *Central and peripheral pressures in one subject (mmHg)*

	Low Sys	**High Sys**	**Low Dia**	**High Dia**	**MAP**
Central	110	120	69	72	85
Peripheral	120	132	64	68	—

Group Data: Supplementary Table 1 provides data from all 92 participants in whom sphygmomanometry was also performed. Table 3 presents the mean ± SD of lower and higher systolic and diastolic pressures obtained from central and peripheral plethysmometric measurements and from sphygmomanometry.

Table 3. *Mean ± SD of systolic and diastolic pressures by method (n = 92)*

n = 92	**Central**				**Peripheral**				**Sphygmomanometry**			
	Low Sys	High Sys	Low Dia	High Dia	Low Sys	High Sys	Low Dia	High Dia	Low Sys	High Sys	Low Dia	High Dia
Mean (mmHg)	109	120	68	74	116	128	66	72	109	117	72	79
SD (mmHg)	14	14	12	12	15	15	12	11	14	15	12	11

Agreement Between Methods: Mean differences (plethysmometry – sphygmomanometry) and corresponding SDs are shown in Table 4.

Table 4: *Mean difference & SD between plethysmometry and sphygmomanometry (mmHg)*

	Central Plethysmometry				**Peripheral Plethysmometry**			
	Low Sys	High Sys	Low Dia	High Dia	Low Sys	High Sys	Low Dia	High Dia
Mean difference (Bias), (mmHg)	−0.05	2.37	−3.43	−4.90	6.93	10.57	−5.79	−6.79
SD (mmHg)	7.32	7.53	9.40	9.37	8.12	8.31	9.68	9.40

AAMI Validation: According to the Association for the Advancement of Medical Instrumentation (AAMI) criteria for device validation, the mean bias must be < 5 mmHg and the SD of differences < 8 mmHg. From Table 4, the low and high systolic pressures satisfy this criterion, whereas the diastolic pressures do not. Criterion 2 of the AAMI standard, which requires three paired measurements per subject, was not assessed because plethysmometry was performed only once. This single measurement was considered sufficient to capture the inherent short-term variability in blood pressure—the core hypothesis of the study.

BHS Grading: The British Hypertension Society (BHS) protocol specifies that, for grade A performance, at least 60 % of readings should differ by ≤ 5 mmHg from the reference, with corresponding thresholds of 50 % and 40 % for grades B and C, respectively. Table 5 shows

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that plethysmometric estimates achieved grade B for both low and high systolic pressures and grade C for diastolic pressures.

Table 5. *BHS grading of agreement between central plethysmometric and sphygmomanometric pressures*

Parameter	≤ 5 mmHg (%)	≤ 10 mmHg (%)	≤ 15 mmHg (%)	Grade
Low Sys	55	78	98	B
High Sys	55	82	96	B
Low Dia	43	75	86	C
High Dia	40	72	86	C

Average Pressures: Average systolic and diastolic pressures were computed as the within-subject mean of lower and upper limits for each individual, then aggregated across participants; the resulting group means for average systolic and diastolic pressures are presented in Table 5.

Table 5. *Group means (n = 92) of average systolic and diastolic pressures from central and peripheral plethysmometry and sphygmomanometry (mmHg)*

Method	Mean Systolic ± SD	Mean Diastolic ± SD
Central plethysmometry	115 ± 14	71 ± 12
Peripheral plethysmometry	122 ± 15	69 ± 11
Sphygmomanometry	114 ± 15	76 ± 11

Frequency distribution of the average pressures for plethysmometry (central and peripheral) and sphygmomanometry are shown in figure 3a. The bin-width for the frequency distribution was arrived at using the Freedman–Diaconis rule (Bin width = 2*(interquartile range)/cube root of sample size).

Figure 4: *Frequency distribution of average systolic and diastolic pressures from plethysmometry (central and peripheral) and sphygmomanometry*

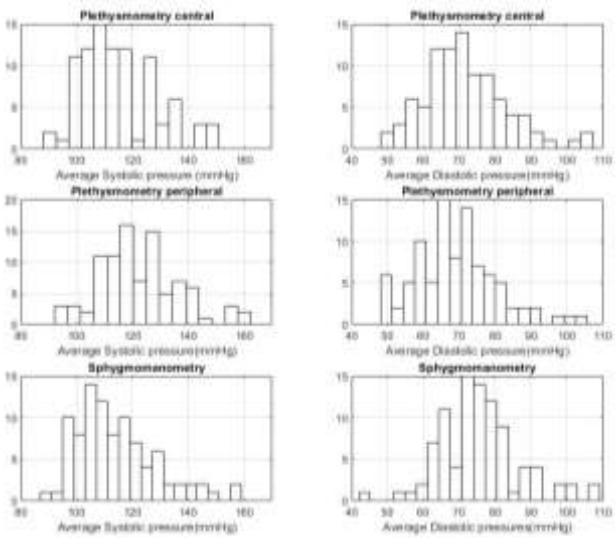

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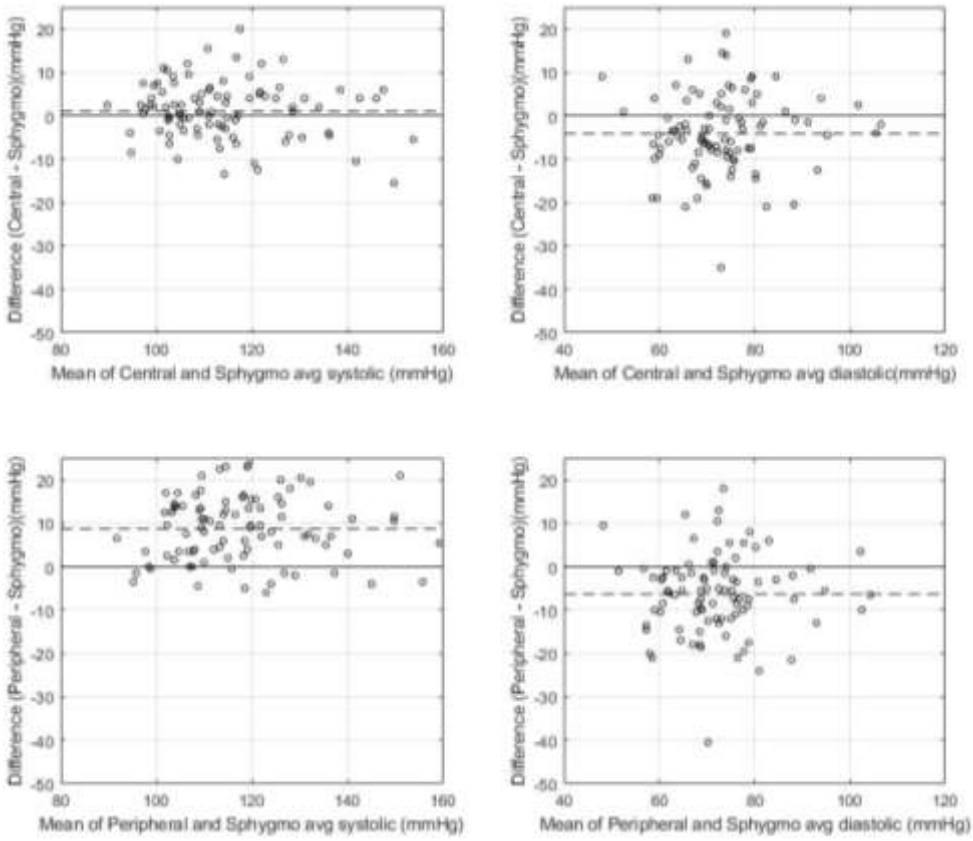
Table 6. *Mean ± SD of differences (Plethysmometry – Sphygmomanometry) for average pressures (mmHg)*

	Central pressures		Peripheral pressures	
	Bias (Mean of differences)	SD	Bias (Mean of differences)	SD
Average Systolic	1.18	6.66	8.82	7.46
Average Diastolic	−4.17	9.07	−6.35	9.25

Here again, if AAMI criterion 1 is applied, the device passes the test for central systolic pressure. For the other pressures, either the bias is more than 5 mmHg and/or the SD is more than 8 mmHg.

Notably, the mean bias for central average diastolic pressure is −4.17 mmHg, reflecting lower plethysmometric diastolic estimates relative to sphygmomanometry. Given published reports that sphygmomanometry tends to overestimate diastolic pressure**,** the observed negative bias may not necessarily indicate inaccuracy of the test device. This should be explored further with comparison with invasive arterial BP measurement.

Figure 5: *Bland Altman plots comparing average Plethysmometric pressures (Central and Peripheral) with Sphygmomanometry*



Percent Agreement by BHS Criteria: The proportions of participants whose absolute differences between central plethysmometry and sphygmomanometry fall within ≤5, ≤10, and ≤15 mmHg are summarized in Table 7.

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Table 7. *Comparison of average systolic and diastolic pressures: central plethysmometry vs sphygmomanometry (BHS criteria)*

Parameter	≤ 5 mmHg (%)	≤ 10 mmHg (%)	≤ 15 mmHg (%)	BHS grade
Average systolic	61	86	98	A
Average diastolic	40	76	89	C

Applying BHS grading to average central pressures indicates good agreement for systolic (grade A) but weaker agreement for diastolic (grade C). This pattern is consistent with the AAMI criterion-1 analysis in Table 6.

Table 8 gives 5 experiments in which either the systolic difference or diastolic difference between Plethysmometry (central) and sphygmomanometry is very high.

Table 8: *Plethysmometric (central) and sphygmomanometric pressures in five subjects where there is over or underestimation of systolic or diastolic pressures by sphygmomanometry:*

		Plethysmometry Central pressures (mmHg)				Sphygmomanometry pressures (mmHg)			
		Low sys	High sys	Low dia	High dia	Low sys	High sys	Low dia	High dia
1	Systolic underestimation	123	132	66	71	106	109	71	80
2	Systolic underestimation	112	125	60	68	101	105	65	70
3	Systolic overestimation	139	145	104	107	155	160	107	108
4	Diastolic underestimation	110	120	82	85	122	130	59	70
5	Diastolic overestimation	110	119	70	74	112	118	89	97

The raw data or graphs in the figures that follow demonstrate the validity of plethysmometric measurements for each of the 5 experiments in Table 8.

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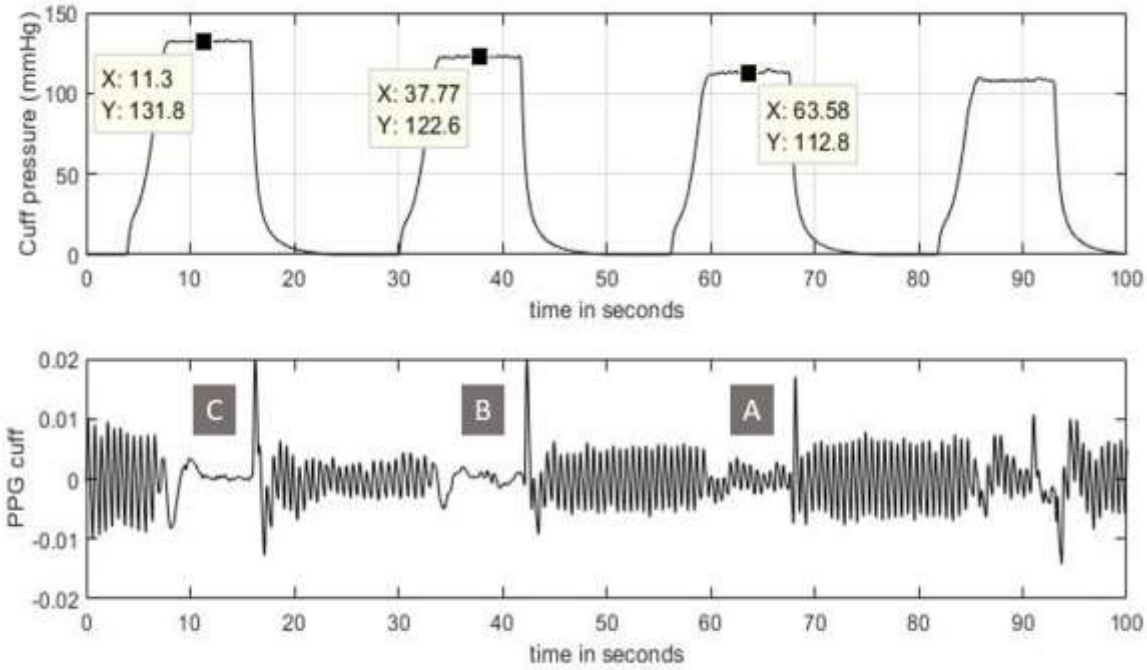
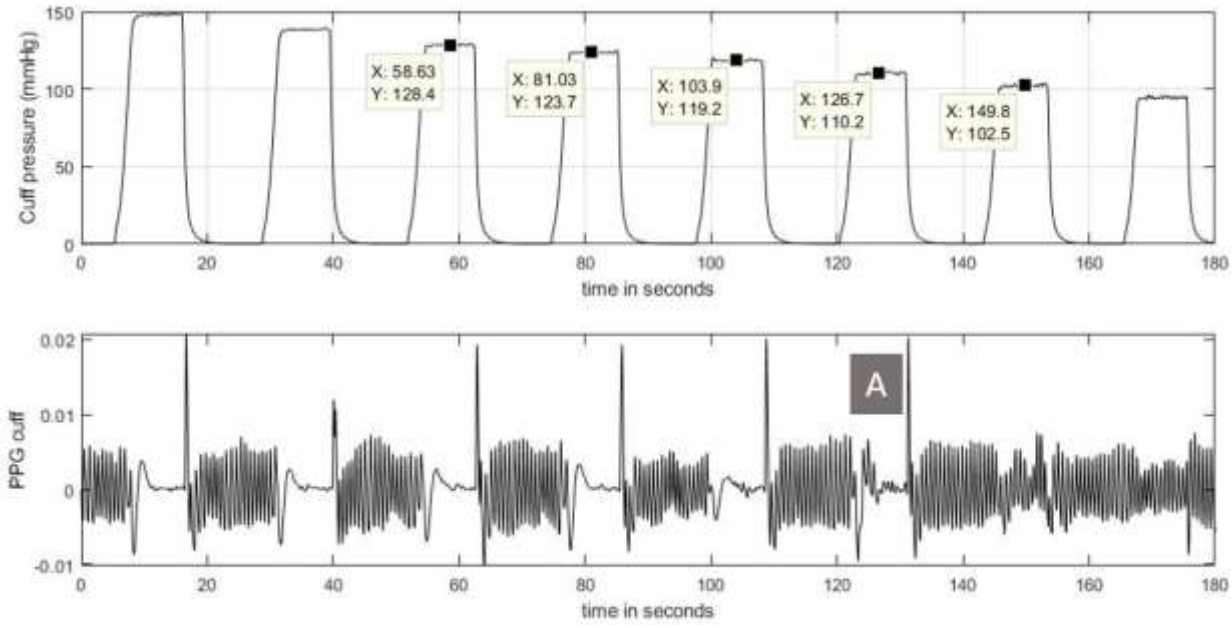
Figure 6: *Raw data for experiment 1 of Table 8:*



The highest systolic pressure recorded in 4 sphygmomanometric experiments is 109 mmHg (see Table 8). However, in figure 6, note that at point A, at a cuff pressure of ~113 mmHg, all pulses have gone through in the cuffed arm. Even at point B, where the cuff pressure is ~123 mmHg, some pulses are present while some are absent. Only at point C, at a cuff pressure of ~132 mmHg, all pulses are absent.

Figure 7: *Raw data for experiment 2 in Table 8:*



In figure 7, note that at point A, even at a cuff pressure of 110 mmHg, pulses are present, whereas the highest systolic pressure recorded in 4 repeats of sphygmomanometry is 105 mmHg (Table 8, experiment no 2).

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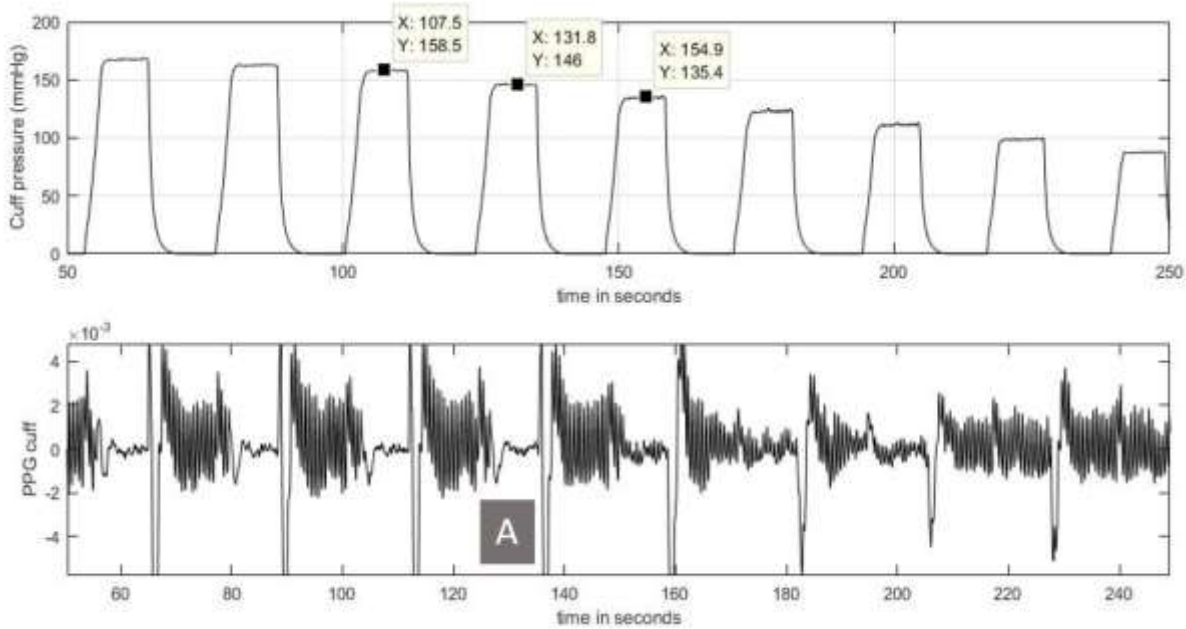
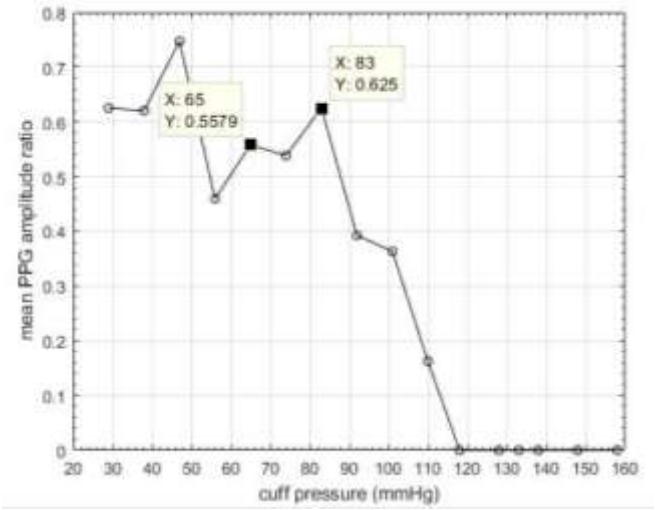
Figure 8: Raw data for experiment 3 in Table 8:



From Table 8, experiment 3, the lowest sphygmomanometric systolic estimate is 155 mmHg. However, note in figure 8, that at point A, all pulses are absent even at a cuff pressure of 146 mmHg. This is an example of over-estimation of systolic pressure by sphygmomanometry.

Figure 9: Plot of mean PPG amplitude ratio versus cuff-pressure for experiment 4 in Table 8.



Note in figure 9, that PPG amplitude is restored at a cuff pressure of 83 mmHg while sphygmomanometry reports highest diastolic in 4 estimates as 70 mmHg. This is an example of under-estimation of diastolic BP by sphygmomanometry.

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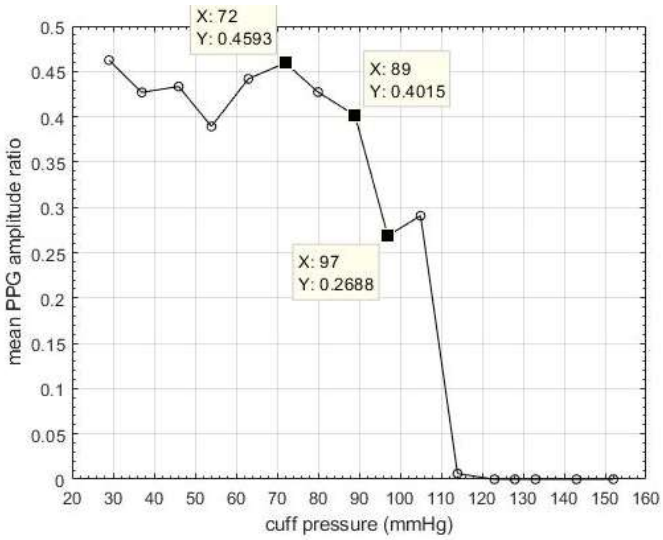
Figure 10: *Plot of mean PPG amplitude ratio versus cuff pressure at plateau for experiment 5 in Table 8.*



Note in Table 8 experiment 5, that the diastolic pressure range from 4 estimates of sphygmomanometry is 89 – 97 mmHg. However, Figure 10 shows that at 97 mmHg, PPG amplitude is only about half of maximum amplitude. Even at 89 mmHg amplitude is not fully restored. Plethysmometry reports central diastolic pressure range as 70 - 74 mmHg at which PPG amplitude restoration is complete. This experiment is an example of over-estimation of diastolic BP by sphygmomanometry.

Comparisons were also made with sphygmomanometry by averaging systolic and diastolic pressures from all 4 repeats, instead of just the lowest and highest values as done in the preceding sections. The data is similar and are given in Supplementary Tables 2a and 2b.

Rationale for comparison of BP Classification for validation: It is inherently challenging for new blood-pressure measurement devices to meet AAMI and BHS validation criteria when comparisons are made directly with sphygmomanometric absolute values of systolic and diastolic pressures. The difficulty arises because the accuracy of sphygmomanometry itself is disputed. Using it as a reference standard may therefore lead to misclassification of an accurate device as inaccurate—as appears to be the case for plethysmometry.

If, however, one assumes that 'classification' of blood pressure based on repeated sphygmomanometric measurements within an individual is reasonably accurate—since diagnostic cut-offs for hypertension already incorporate the systematic errors inherent in sphygmomanometry—then comparing BP classification, rather than absolute pressure values, would provide a more meaningful validation of a new device. Accordingly, a scheme for BP classification using plethysmometry (central pressures) was developed.

Central Plethysmometric Pressures in Individuals Without Known Hypertension: In 77 individuals without a prior diagnosis of hypertension and in whom average sphygmomanometric pressures were below 140/90 mmHg (see Table 1a), average systolic, diastolic, mean arterial (MAP), and pulse pressures were calculated from the lower and upper limits of each parameter obtained from plethysmometry. Pulse pressure (PP) was computed as the difference between average systolic and diastolic pressures for each individual. The group mean and standard deviation (SD) for the 77 experiments are presented in Table 9.

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Table 9. *Mean ± SD of plethysmometric (central) pressures in individuals without a previous history of hypertension and normotensive as per sphygmomanometry (n = 77)*

Parameter	Mean (mmHg)	SD (mmHg)
Average systolic	111	10
Average diastolic	69	9
Average MAP	84	8
Average pulse pressure	42	10

A spectrum of pressures was then described ranging from Mean + 0.5 SD to Mean + 2.5 SD with the values shown in Table 9. Various thresholds in the spectrum are shown in Table 10.

Table 10: *Thresholds for classification of central Blood pressure from plethysmometry*

All values in mmHg	Average systolic pressure	Average diastolic pressure	Average MAP	Average pulse pressure
Mean (n = 77)	111	69	84	42
Mean + 0.5 SD	116	74	88	47
Mean + SD	121	78	92	52
Mean + 1.5 SD	126	83	96	57
Mean + 2 SD	131	87	100	62
Mean + 2.5 SD	136	92	104	67

BP classification for individual subjects: Scores were ascribed to each subject's pressures depending on where it occurred in the spectrum ranging from (sample mean + 0.5 SD) to (sample mean + 2.5 SD)) for each of the 4 pressure parameters (see Table 10). The scoring system is given in Table 11. The scoring system was arrived at by iteration so as to get maximum sensitivity and specificity for hypertension diagnosis by plethysmometry, as compared to 4 repeats of video/audio recorded sphygmomanometry.

Table 11: Scoring system for BP classification by Plethysmometry:

Score	For each of Systolic, Diastolic, MAP and PP
0	Subject's pressure ≤ mean + 0.5 SD
1	mean + 0.5 SD < Subject's pressure ≤ mean + SD
2	mean + SD < Subject's pressure ≤ mean + 1.5 SD
3	mean + 1.5 SD < Subject's pressure ≤ mean + 2 SD
4	mean + 2 SD < Subject's pressure ≤ mean + 2.5 SD
5	mean + 2.5 SD < Subject's pressure

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The individual scores for the 4 pressure parameters were summed to get 'Total score'. Total score can be in the range from 0 to 20. A person was classified as hypertensive if the total score was more than 9 and as normotensive if the total score was less than 6. For scores between 7 and 9, if certain criteria were met, the diagnosis was Hypertension. The diagnostic criteria were derived by iteration to get the best sensitivity and specificity for this study as well as another study where plethysmometry was compared with diagnosis by 24-hr Ambulatory Blood pressure monitoring (ABPM).

BP classification for the 92 subjects of this study was also made with average sphygmomanometric pressures with the usual cut-off of ≥ 140/90 mmHg. The sphygmomanometric pressures were averaged from the highest and lowest of 4 measurements. The contingency table for comparison of hypertension diagnosis by both methods is given in Table 12:

Table 12: *Comparison of Hypertension diagnosis with plethysmometric pressures and the scoring system described in Table 11 with sphygmomanometric diagnosis:*

		Sphygmomanometry (average of 4 repeats)	
		HT (11)	NT (81)
Plethysmometry (central)	HT (13)	**10** (True positive)	**5** (False positive)
	NT (79)	**1** (False negative)	**76** (True negative)

Pressure data for the 5 false positive and 1 false negative subjects shown in the contingency table is given in Supplementary Table 3.

Metrics for the above contingency table is given in Table 13.

Table 13: *Sensitivity and Specificity of Plethysmometry as a tool for diagnosis of hypertension as compared to diagnosis made by average of the highest and lowest estimates from 4 carefully performed sphygmomanometric estimations*

Sensitivity	Specificity	Positive predictive value	Negative predictive value
0.91	0.94	0.67	0.99

When sphygmomanometric pressures were averaged from all 4 measurements, instead of the highest and lowest as done for Table 12, there was 1 less false positive. The metrics for this comparison are given in Table 14.

Table 14: *Sensitivity and Specificity of Plethysmometry when compared with sphygmomanometric pressures averaged from all 4 measurements*

Sensitivity	Specificity	Positive predictive value	Negative predictive value
0.92	0.95	0.73	0.99

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Discussion: Sphygmomanometry is traditionally considered the reference standard for validating new blood pressure (BP) measuring devices. However, its accuracy has long been debated (3-5). The inaccuracy of sphygmomanometry may arise not only from noisy environments or impaired hearing by the observer but also from fundamental limitations of the auscultatory method itself. In the present study, sphygmomanometry was performed under highly controlled conditions—in a sound-proof room with simultaneous video recording of the mercury column and Korotkoff sounds. Each subject underwent four measurements, two before and two after plethysmometry. Two observers independently reported systolic and diastolic pressures; if their readings differed by more than 2 mmHg, a third observer acted as arbiter. Despite these precautions, Korotkoff sounds did not always appear and disappear precisely at systolic and diastolic pressures, suggesting that additional factors such as arterial wall vibration influence sound generation.

Consequently, when a new device is compared directly with sphygmomanometry in terms of absolute pressures, an objectively more accurate instrument may appear inaccurate merely because the reference method itself is flawed. The plethysmometric device developed in this study illustrates this paradox.

Given that sphygmomanometry has been in clinical use for more than a century and is employed by millions of healthcare professionals worldwide, the diagnostic thresholds for hypertension are necessarily comprehensive, having implicitly accommodated the systematic errors of this method.

A more meaningful validation approach therefore, would be comparison of BP classification outcomes, rather than absolute pressure values, between the test device and sphygmomanometry. The underlying assumption is that averaged pressures from multiple sphygmomanometric measurements in the same individual provide a reasonably accurate classification of hypertensive versus normotensive status, even if the absolute values contain bias.

Using this framework, the plethysmometric method—which includes the patented device, the proprietary algorithm for pressure estimation, and the diagnostic scoring scheme—achieved a sensitivity and specificity more than 0.9 for hypertension detection. A single plethysmometric recording thus substitutes for multiple sphygmomanometric measurements that require trained observers and ideal acoustic conditions.

The negative predictive value (NPV) was 0.99, indicating that individuals identified as normotensive by plethysmometry are almost certainly normotensive. The positive predictive value (PPV) of 0.67 suggests that some cases may be over-called as hypertensive, but this is desirable for a screening tool, where missing true hypertension is of greater concern than false positives.

Importantly, the five individuals labelled hypertensive by plethysmometry but normotensive by sphygmomanometry (so-called *false positives*) are likely to represent cases of masked hypertension—a condition in which BP is normal during clinic measurements but elevated outside the clinical setting. Masked hypertension can only be detected through out-of-office monitoring, such as home BP recordings or 24-hour ambulatory BP monitoring (ABPM). Currently, subjects with normal office BP are rarely evaluated further; hence, masked HT

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remains undiagnosed. The ability of plethysmometry to identify such cases highlights its potential clinical advantage.

Future Directions Further studies should compare plethysmometric pressures with intra-aortic and intra-arterial measurements to confirm that central plethysmometric values correspond to aortic pressures and peripheral plethysmometric values to radial or brachial pressures. Preliminary pilot experiments in 16 subjects showed mixed correspondence—some cases matched radial intra-arterial pressures to peripheral plethysmometry, whereas in others the central plethysmometric readings were closer. This may be due to over- or under-damping of the intra-arterial catheter system. Although fast flush-tests were performed, they were not recorded; hence, the degree of damping could not be confirmed. These experiments will be repeated with confirmation of optimal natural frequency and damping of the catheter system.

Validation should also include comparison of plethysmometric diagnosis with 24-hour ABPM, particularly in individuals classified as hypertensive by office BP, and should extend to special populations such as pregnant women, children, neonates, and patients in shock or critical illness.

Conclusion: The newly patented plethysmometric BP device is an accurate and reliable tool for measuring blood pressure. Unlike conventional devices, it provides not only single-point systolic and diastolic values but also their ranges over time, thereby quantifying ultrashort-term BP variability (BPV). It also yields the mean arterial pressure and pulse pressure. The usefulness of the BPV parameter will be evaluated in future.

The diagnostic algorithm for HT by the new method incorporates systolic, diastolic, mean arterial, and pulse pressures. Although larger normative datasets are required to refine diagnostic thresholds, results from the present cohort of 92 subjects (thresholds derived from a normative sample of 77) show high diagnostic accuracy when compared with four sphygmomanometric readings taken under ideal conditions.

Moreover, plethysmometry produces objective digital records from which systolic and diastolic determinations can be verified—an advantage not shared by either sphygmomanometry or Oscillometry.

In summary, plethysmometry presents a promising advancement in BP measurement and hypertension diagnosis, combining accuracy and objective verification in a single, easy-to-use platform.

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Patents:

Sathya Subramani and Benjamin Jebaraj, 2023; Non-Invasive system and method for measuring blood pressure variability;

4. Indian Patent No: 428809 (Indian patent office)
5. US Patent No: US 11,723,543 B2, (United States Patent and Trademark Office)
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